Samuel Degen

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EDUCATION

University of California Los Angeles (UCLA) *9 / 2023–present*

B.S. Physics ◇ GPA: 3.98 (overall); 4.00 (major) ◇ 16 graduate physics classes

Advisors: Zvi Bern & Mikhail Solon (Scattering Amplitudes)

University of Colorado Boulder *2021–2023*

High School Concurrent Enrollment ◇ GPA: 4.0 (overall)

PUBLICATIONS

1. S. Wang, **S. Degen** and H. Liang, "Functional methods for quantum thermodynamics", *revision submitted to Physical Review B*, 2026. [2605.31112]
2. Z. Bern, **S. Degen,** E. Herrmann, M.P. Solon and A.M. Wolz, "Environmental Effects in Post-Minkowskian Dynamics: Effective Field Theory, Feynman Rules, and Ward Identities for Compact Objects in Relativistic Fluids", *submitted to Physical Review D*, 2026. (*primary contributor; authors alphabetical*) [2609.20742]
3. S. Wang, **S. Degen** and H. Liang, "Functional-Renormalization-Group Density-Matrix Functional Theory: Benchmarking on the Lipkin–Meshkov–Glick model in the Normal Phase", *in preparation*, 2026. [2611.XXXXX]

HONORS

Rhodes Scholarship Finalist (interviewing in mid-November) *10 / 2026*

Nominated by UCLA for Marshall, Rhodes, and Churchill Scholarships *Fall 2026*

UCLA Undergraduate Research Center–Sciences Summer Program *Summer 2026*

Goldwater Scholarship (Featured in UCLA College Interview and UCLA Physics News) *2026 - 2027*

- 1 of 2 UCLA students in 2026 awarded national research scholarship

American Physical Society Steven Chu Award for Best Undergraduate Research (1st prize) (Featured on the APS website and UCLA Physics News) *10 / 2025*

UCLA Undergraduate Research Scholars Program *2025 - 2026*

Mani L. Bhaumik Institute for Theoretical Physics Summer Research Fellowship *8 - 9 / 2025*

Friends of UTokyo Inc. Global Leadership Award (report) *2025*

University of Tokyo Research Internship Program (UTRIP) (final report) *6 - 7 / 2025*

- Selected from 1,149 applicants for summer research and culture program in Tokyo

Perimeter Institute for Theoretical Physics' PSI START Satellite Program (interview) *5 / 2025*

UCLA Department of Physics Summer Undergraduate Research Fellowship *Summer 2024*

University of Science and Technology of China Future Scientist Exchange Program (*declined*) *Summer 2026*

RESEARCH EXPERIENCE

Particle Theory (Scattering Amplitudes) @ UCLA
8 / 2025–present

Advisors: Zvi Bern & Mikhail Solon

- Developed a hydrodynamic effective field theory for compact objects in general-relativistic inviscid fluids within a unified, power-counted diagrammatic expansion by drawing connections between gravitational environmental effects and jet-quenching physics in QCD.
- Proved gauge consistency of graviton–phonon Feynman rules using generalized on-shell Ward identities in a Lorentz-broken theory, developing the framework needed for order-by-order in Newton's G (post-Minkowskian) calculations of environmental effects on binary dynamics.
- Showed that observable amplitudes in weakly self-gravitating environments factorize into vacuum-gravity and fluid sectors, allowing phonon–graviton kinetic mixing to be retained and place environmental effects under the same perturbative control as vacuum gravitational scattering calculations.
- Demonstrated power of EFT by recovering the state-of-the-art relativistic dynamical friction force for linear motion from a tree-level single-phonon emission amplitude.
- Youngest member and only undergraduate in the European Research Council's GWSky collaboration, contributing to a new international effort on precision theory for next-generation gravitational-wave astronomy; serving as a coordinator for the *EFT for Environments* project.
- Presented this work at the GWSky Sexten Meeting, IHES Summer School, California Amplitudes Meeting, and UCLA Amplitudes Journal Club.

Supported by UCLA URCSSP, Mani L. Bhaumik Institute Summer Research Fellowship, and UCLA URSP.

Associated Publication: arXiv:2609.20742 [hep-th]

Nuclear Theory @ Quark Nuclear Science Institute, The University of Tokyo
5/2025–present

Advisor: Haozhao Liang

- Applying functional renormalization group-aided density functional theory (FRG-DFT) to exactly solvable quantum many-body systems, with the goal of benchmarking nonperturbative first-principles approaches to nuclear density functionals.
- Deriving and truncating flow-equation hierarchies for ground-state energies and correlation functions, enabling controlled comparisons between FRG-DFT, perturbation theory, and exact thermodynamics.
- Developing numerical implementations to solve coupled flow equations and test the accuracy of approximate truncations across interaction strengths and densities.
- Co-authored a manuscript on benchmarking FRG-DFT against exact thermodynamics for the single-site Bose-Hubbard model, with broader applications to strongly correlated fermions and nuclear many-body theory.
- Presented related work at the University of Tokyo, APS Far West Section Meeting, and APS Global Physics Summit 2026; awarded 1st prize of the APS Steven Chu Award for Best Undergraduate Research for this project.

Supported by UTRIP 2025 and FUTI Global Leadership Award.

Associated Publication: arXiv:2605.31112 [cond-mat.quant-gas]

Plasma Theory and Machine Learning @ UCLA
4 / 2024–1 / 2025

Advisor: E. Paulo Alves

- Developed analytic and machine-learning methods to study time-dependent nonthermal particle acceleration in relativistic astrophysical jets.
- Formulated an inverse-modeling approach to distinguish physical acceleration histories from degenerate solutions that reproduce the same observed particle spectrum.
- Demonstrated limitations of simple energy-dependent acceleration models and identified more flexible time-dependent descriptions consistent with simulated jet dynamics.
- Presented results in an oral contribution at the 66th Annual Meeting of the APS Division of Plasma Physics.

Supported by UCLA Physics Summer 2024 Research Fellowship.

Condensend Matter Experiment @ UCLA
1 / 2024–3 / 2024

Advisor: B.C. Regan

- Developed an undergraduate-level pedagogical treatment of the Feynman checkerboard model, emphasizing its connection to propagators and relativistic quantum mechanics.

PRESENTING & TALKS (◊ = INVITED, ★ = AWARDED)

"The Astounding Success of Effective Theories: From Quarks and Gluons to Black Hole Mergers"

◊ Guest Lecture on Theoretical Physics, UCLA Olga Radko Endowed Math Circle (ORMC) *[planned November 22, 2026]*

"Combining Human and Artificial Intelligence for Practical Functional Renormalization Group Flows"

- Participant talks, California AI Amplitudes School and Workshop, UCLA *[planned November 21, 2026]*

"Ward Identities for Compact Objects in Relativistic Fluids"

- Weekly Seminar, Amplitudes Journal Club, UCLA *[planned November 5, 2026]*

"An Effective Field Theory Framework for Compact Objects in General-Relativistic Inviscid Fluids"

- Summer Undergraduate Research Showcase, UCLA *August 26, 2026*

"Synergistic Approaches to Precision Environmental Effects in Gravitational Waveform Modeling" (with Francisco Duque)

- EFT For Environment Project Coordinators' Report, GWSky Workshop - Precision Gravitational Wave Astronomy: From Theory to Discovery, Sexten Center for Astrophysics Riccardo Giacconi; Sexten, Italy *July 29, 2026*

"Progress on Computing the Dynamical Friction Force at next order in G_N"

- GWSky Online Monthly Meeting *July 20, 2026*
- Flash Talks, GWSky Workshop - Precision Gravitational Wave Astronomy: From Theory to Discovery, Sexten Center for Astrophysics Riccardo Giacconi; Sexten, Italy *July 28, 2026*

"Gravitational Drag from Phonon Emission"

- Weekly Seminar, Amplitudes Journal Club, UCLA *April 23, 2026*
- California Amplitudes Meeting 2026 (Davis), UC Davis *May 30, 2026*
- Cosmological Correlators Summer School Gong Show, Institut des Hautes Études Scientifiques (IHES), France *July 10, 2026*

"Effects of Black Hole Environments on Gravitational Wave Signals"

- California Amplitudes Meeting 2025 (LA), UCLA *November 8, 2025*
- Undergraduate Research & Creativity Showcase, UCLA *May 19, 2026*

"Towards a Nonperturbative Many-Body Nuclear Theory: Benchmarking FRG-DFT with the One-Dimensional Fermi Gas"

- UTRIP Final Presentation, The University of Tokyo *August 5, 2025*

★ Nuclear Physics Oral Contributions (L04.00005), APS Far West Section Meeting, UC Santa Cruz. *Steven Chu Award (1st)* *October 11, 2025*

- Nuclear Theory Oral Contributions, APS Global Physics Summit 2026; Denver, Colorado *March 17, 2026*

"Physics Without Borders: Nuclear Theory and Global Perspectives from a Summer in Tokyo"

◊ Special Seminar, Physics Summer Research Experience for Undergraduates (REU), UCLA *August 15, 2025*

"From Finite Groups to Lie Algebras: Symmetries and Representations in Physics"

- Perimeter Institute's PSI START Satellite Program, Bishop's University; Quebec, Canada *May 29, 2025*

"Data-driven statistical model of nonthermal particle acceleration by the kink instability in relativistic jets"

- Department of Physics Summer Research Talks, UCLA *August 22, 2024*

- Plasma Astrophysics Oral Contributions (NO05.00014), 66th Annual Meeting of the APS Division of Plasma Physics; Atlanta, Georgia — *October 9, 2024*

MEETINGS & WORKSHOPS

QCD Meets Gravity 2026 — *[planned 12 / 2026]*
Humboldt-Universität zu Berlin, Germany

California AI Amplitudes School and Workshop — *[planned 11 / 2026]*
Mani L. Bhaumik Institute for Theoretical Physics, UCLA

GWSky Workshop - Precision Gravitational Wave Astronomy: From Theory to Discovery — *7 / 2026*
Sexten Center for Astrophysics Riccardo Giacconi, Italy
(Coordinator of the "EFT for Environments" project, organizing discussion sessions and presenting on synergestic next steps for the collaboration)

2026 IHES Summer School - Cosmological Correlators — *7 / 2026*
Institut des Hautes Études Scientifiques (IHES), France
(Selected participant; advanced international school primarily for graduate students and postdocs)

California Amplitudes Meeting 2026 (Davis) — *5 / 2026*
Center for Quantum Mathematics and Physics (QMAP), UC Davis

APS Global Physics Summit 2026 — *3 / 2026*
Denver, Colorado

Accelerating Math and Theoretical Physics with AI — *3 / 2026*
Institute for Pure and Applied Mathematics (IPAM), UCLA

GWSky Kick-off Meeting — *12 / 2025*
Max Planck Institute for Gravitational Physics (Albert Einstein Institute–AEI), Potsdam, Germany

California Amplitudes Meeting 2025 (LA) — *11 / 2025*
Mani L. Bhaumik Institute for Theoretical Physics, UCLA

2025 Annual Meeting of the APS Far West Section — *10 / 2025*
UC Santa Cruz

2025 Southern California Strings — *10 / 2025*
Mani L. Bhaumik Institute for Theoretical Physics, UCLA

California Amplitudes Meeting 2025 (Davis) — *5 / 2025*
Center for Quantum Mathematics and Physics (QMAP), UC Davis

66th Annual Meeting of the APS Division of Plasma Physics — *10 / 2024*
Atlanta, Georgia

GRADUATE COURSEWORK (18 TOTAL)

At UCLA: Statistical Physics, Quantum Mechanics I, Quantum Mechanics II, Particle Physics (Standard Model), Particle Physics (QCD and Higgs), Beyond the Standard Model, Quantum Field Theory I, Quantum Field Theory II, Quantum Field Theory III, Advanced Quantum Field Theory, Mathematical Physics (Lie Theory), Gravitational Waves & the Double Copy, Solid State Physics, Topological Quantum Field Theory, String Theory Journal Club, Teaching College Physics, Lie Groups & Lie Algebras

At CU Boulder: Gravitational Theory, Measure-Theoretic Probability

Final Papers:

- PHYS 242C Topological Quantum Field Theory: S. Degen, "An Introduction to 3D Gravity from TQFTs" — *6 / 2025*
- MATH 229A Lie Groups & Lie Algebras: S. Degen, "Tensor Products of Representations of Lie groups" — *10 / 2024*

Final Presentations:

- PHYS 226D Beyond the Standard Model: "An Introduction to the Dark Photon" *December 4, 2025*
- APPM 6560 Measure-Theoretic Probability: "Convex Compactness and its Applications" *May 1, 2023*

COMMUNITY & SERVICE

Guest Lecturer, UCLA Olga Radko Endowed Math Circle (ORMC) [*planned November 22, 2026*]

- Invited to deliver a 90-minute guest lecture followed by Q&A on theoretical physics to gifted pre-college mathematics students in Los Angeles' largest math circle, communicating advanced ideas to a non-specialist audience while demonstrating that serious scientific research is accessible to young students

Peer Mentor, Division of Physical Sciences, UCLA *2025–present*

- Provided first-year physical science students at UCLA guidance on coursework, professional development, fellowships, and early research opportunities, especially for those interested in theoretical physics

Member, UCLA String Theory and Amplitudes Journal Clubs *1 / 2025–present*

- First undergraduate member and presenter of these journal clubs

Professional Events Chair, Society of Physics Students, UCLA *2025*

- Developed extensive resources for summer research opportunities and course planning

Member, UCLA Plasma Theory Journal Club *4 / 2024 - 1 / 2025*

- Presented recent papers to theoretical plasma PhD students and faculty
- Discussed theory and application of novel analytic and machine learning methods presented in the club to current projects in the UCLA Plasma Theory group